POC

Jira → GitLab Sync Service

Automated replication of Jira Stories, Bugs & Tasks to GitLab milestones using a .NET 8 Windows Service — Proof of Concept

.NET 8	Jira REST API v3	GitLab REST API v4	Windows Service

Raghubir Singh · June 2026

Problem Statement

Three challenges driving the need for this sync service

01 Limited Jira Access

Entire team cannot see work items

Stories and Bugs are created in Jira, but only a few members have licences. Developers and QA who work in GitLab daily have zero visibility into sprint scope, ticket details, or planned work — creating a blind spot across the team.

02 Disconnected Tools

Code in GitLab, work tracking in Jira

All code repositories live in GitLab which the whole team can access, but sprint planning and stories remain siloed in Jira. Context-switching between two systems makes it hard to link commits to requirements or track sprint progress.

03 Manual Excel Tracking

Team managing tasks and status in spreadsheets

Without a shared system, members maintain tasks, assignments and work status in Excel files. This causes version conflicts, stale data, no sprint-wise tracking, and zero traceability between work items and the code being delivered.

Solution Benefits

What the team gains by syncing Jira to GitLab

👥 Whole Team Visibility

Every member — dev, QA, designer — views sprint tickets in GitLab with no Jira licence needed.

🗂️ Single Source of Truth

Replaces fragmented Excel sheets. Work status, assignees and story points always up to date.

📅 Sprint-wise Tracking

Tickets map to GitLab milestones. Track velocity, completion rate and sprint health.

🔗 Code ↔ Work Traceability

Developers reference ticket IDs in commits — linking code changes to business requirements.

⚡ Zero Manual Effort

Runs automatically on schedule. No copy-pasting, no spreadsheet updates — fully hands-free.

📊 Better Retrospectives

GitLab burndown charts and milestone reports give accurate sprint data for stakeholder updates.

✓ No Jira licence needed ✓ Replaces Excel tracking ✓ Sprint progress in GitLab ✓ Fully automated

Solution Overview

A .NET 8 Windows Service that automatically replicates Jira tickets to GitLab on a configurable schedule

Jira Cloud	→	Sync Service	→	GitLab
Stories · Bugs · Tasks		.NET 8 Windows Service		Issues · Milestones

⏱️ Scheduled Sync

Runs at configured times (e.g. 09:00 IST). Auto-retry on failure.

🎯 Smart Routing

Active milestone today → sprint. No milestone → Backlog.

🔁 Create & Update

New tickets created; existing issues updated and moved to correct milestone.

🏷️ Field Mapping

ID, Description, Assignee, Sprint Dates, Story Points all synced.

🛡️ Error Handling

Global exception handler with per-ticket isolation — one failure won't stop others.

⚙️ Config-Driven

All settings in appsettings.json — schedule, tokens, project key.

Architecture

Layered .NET 8 service with clean separation of concerns

Worker Layer	SyncWorker (BackgroundService)	Timer loop · Retry logic · Schedule mode · Graceful shutdown
Orchestration Layer	SyncOrchestrator	Milestone routing · Create vs Update decision · Error isolation per ticket
Service Layer	JiraService + GitLabService	REST API clients · Basic Auth & Private Token · Pagination · Milestone cache
Model Layer	JiraModels · GitLabModels · SyncTicket	Typed DTOs · ADF → plain-text · Field resolution · SyncResult tracking

Field Mapping

How Jira ticket data maps to GitLab issue fields

Jira Field	GitLab Field	Notes
Story / Bug / Task Key	Issue title prefix [PROJ-42]	Enables duplicate detection on re-sync
Summary	Issue title (after key prefix)	Full summary preserved
Description (ADF format)	Issue description (Markdown)	Atlassian ADF flattened to plain text
Assignee name / email	Assignee (matched by name/email)	Skipped on free-tier GitLab (403 on /users)
Sprint start → end date	Shown in description: From → To	Also drives milestone date matching
Story Points (customfield)	Issue weight field	Tries customfield_10016, 10028, story_points
Issue type (Story/Bug/Task)	GitLab label (story/bug/task)	Configurable via LabelMappings in config

Service Execution Log

Sync run showing 16 tickets created in GitLab Sprint-1 milestone

✓ Sync run complete — Created=16, Updated=0, Skipped=0, Failed=0

```
      Received HTTP response headers after 505.3854ms - 200
info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[101]
      End processing HTTP request after 505.5256ms - 200
info: JiraGitLabSync.Services.GitLabService[0]
      Creating GitLab issue for SCRUM-5
info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[100]
      Start processing HTTP request POST https://gitlab.com/api/v4/projects/82304697/issues
info: System.Net.Http.HttpClient.IGitLabService.ClientHandler[100]
      Sending HTTP request POST https://gitlab.com/api/v4/projects/82304697/issues
info: System.Net.Http.HttpClient.IGitLabService.ClientHandler[101]
      Received HTTP response headers after 663.5037ms - 201
info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[101]
      End processing HTTP request after 663.5634ms - 201
info: JiraGitLabSync.Services.SyncOrchestrator[0]
      Created : [SCRUM-5] 'Screen is taking so much time to load and then timeout issue…' → milestone 'Sprint-1'
info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[100]
      Start processing HTTP request GET https://gitlab.com/api/v4/projects/82304697/issues?search=%5BSCRUM-4%5D&scope=all&per_page=5
info: System.Net.Http.HttpClient.IGitLabService.ClientHandler[100]
      Sending HTTP request GET https://gitlab.com/api/v4/projects/82304697/issues?search=%5BSCRUM-4%5D&scope=all&per_page=5
info: System.Net.Http.HttpClient.IGitLabService.ClientHandler[101]
      Received HTTP response headers after 408.6271ms - 200
info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[101]
      End processing HTTP request after 408.7168ms - 200
info: JiraGitLabSync.Services.GitLabService[0]
      Creating GitLab issue for SCRUM-4
info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[100]
      Start processing HTTP request POST https://gitlab.com/api/v4/projects/82304697/issues
info: System.Net.Http.HttpClient.IGitLabService.ClientHandler[100]
      Sending HTTP request POST https://gitlab.com/api/v4/projects/82304697/issues
info: System.Net.Http.HttpClient.IGitLabService.ClientHandler[101]
      Received HTTP response headers after 665.6504ms - 201
info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[101]
      End processing HTTP request after 665.7054ms - 201
info: JiraGitLabSync.Services.SyncOrchestrator[0]
      Created : [SCRUM-4] 'SQL Connection Timout Error Occuring Occasionally. It happen…' → milestone 'Sprint-1'
info: JiraGitLabSync.Services.SyncOrchestrator[0]
      === Sync run complete - Created=16, Updated=0, Skipped=0, Failed=0 ===
info: JiraGitLabSync.Workers.SyncWorker[0]
      Next sync in 15 minute(s) at ~14:06:54 UTC.
```

Service Execution Log

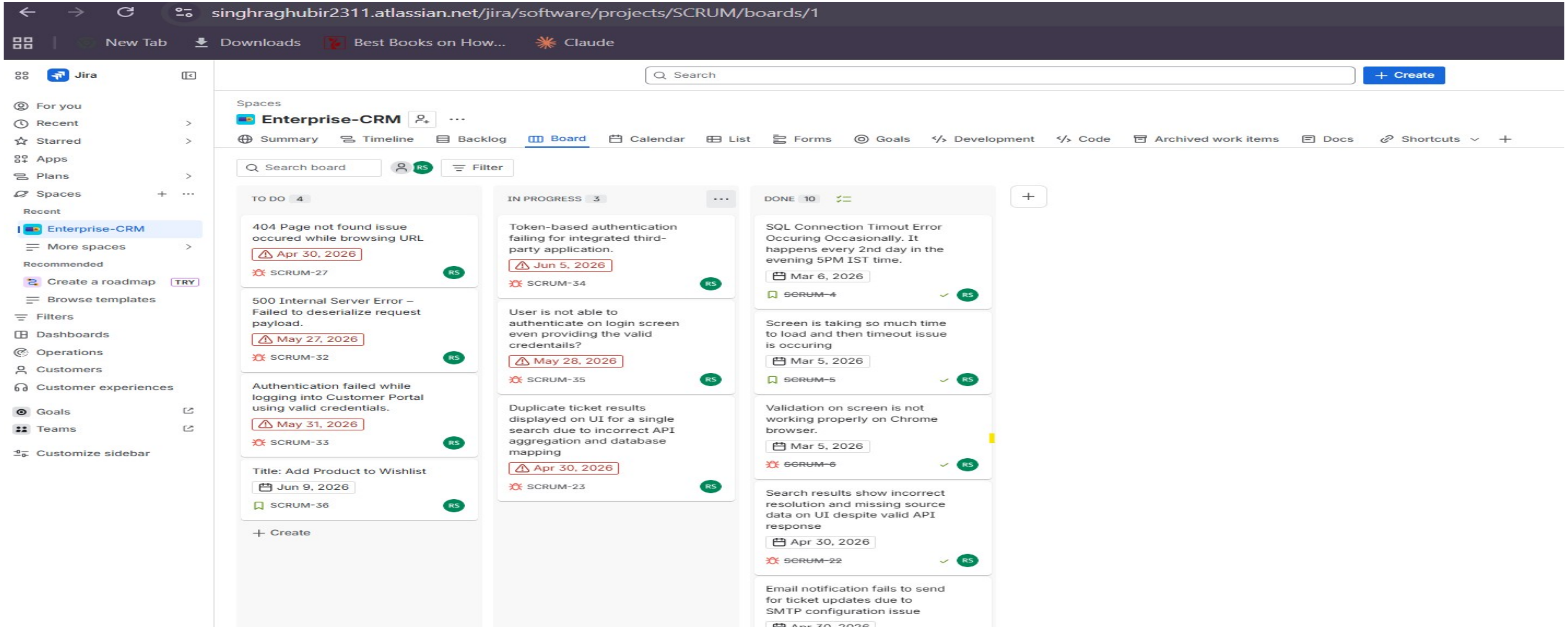
Subsequent run showing update sync — 1 created, 16 updated with no failures

✓ **Sync run complete — Created=1, Updated=16, Skipped=0, Failed=0**

```
      End processing HTTP request after 585.8092ms - 200
info: JiraGitLabSync.Services.GitLabService[0]
      Updating GitLab issue !33 for SCRUM-4 → milestone_id=7433218
info: JiraGitLabSync.Services.GitLabService[0]
      PUT api/v4/projects/82304697/issues/33 body={"title":"[SCRUM-4] SQL Connection Timout Error Occu
 It happens every 2nd day in the evening 5PM IST time. ","description":"## Jira Reference: [SCRUM-4](
browse/SCRUM-4)\r\n\r\n**Type:** Story\r\n**Assignee:** Raghubir Singh\r\n**Sprint Period:** 2026-02-
0\r\n**Priority:** Medium\r\n**Jira Status:** Done\r\n\r\n---\r\n\r\n### Description\r\n\r\nThis was
JOB was running in the backend at 5:00 PM IST and occupying the DB resources. SQL Connection Timeout
 this reason. As a solution, SQL Job is optimized and shifted and rescheduled to odd hours.\r\n","lab
y::medium","milestone_id":7433218,"assignee_ids":[],"weight":null}
info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[100]
      Start processing HTTP request PUT https://gitlab.com/api/v4/projects/82304697/issues/33
info: System.Net.Http.HttpClient.IGitLabService.ClientHandler[100]
      Sending HTTP request PUT https://gitlab.com/api/v4/projects/82304697/issues/33
info: System.Net.Http.HttpClient.IGitLabService.ClientHandler[101]
      Received HTTP response headers after 1287.2385ms - 200
info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[101]
      End processing HTTP request after 1287.3595ms - 200
info: JiraGitLabSync.Services.GitLabService[0]
      PUT response body={"id":191949808,"iid":33,"project_id":82304697,"title":"[SCRUM-4] SQL Connect
curing Occasionally. It happens every 2nd day in the evening 5PM IST time.","description":"## Jira Re
https://gitlab.com/browse/SCRUM-4)\n\n**Type:** Story\n**Assignee:** Raghubir Singh\
info: JiraGitLabSync.Services.SyncOrchestrator[0]
      Updated : [SCRUM-4] !33 'SQL Connection Timout Error Occuring Occasionally. It happen…' → miles
info: JiraGitLabSync.Services.SyncOrchestrator[0]
      === Sync run complete - Created=1, Updated=16, Skipped=0, Failed=0 ===
info: JiraGitLabSync.Workers.SyncWorker[0]
      Next sync in 15 minute(s) at ~16:42:20 UTC.
```

Source: Jira Board

4	To Do
3	In Progress
10	Done

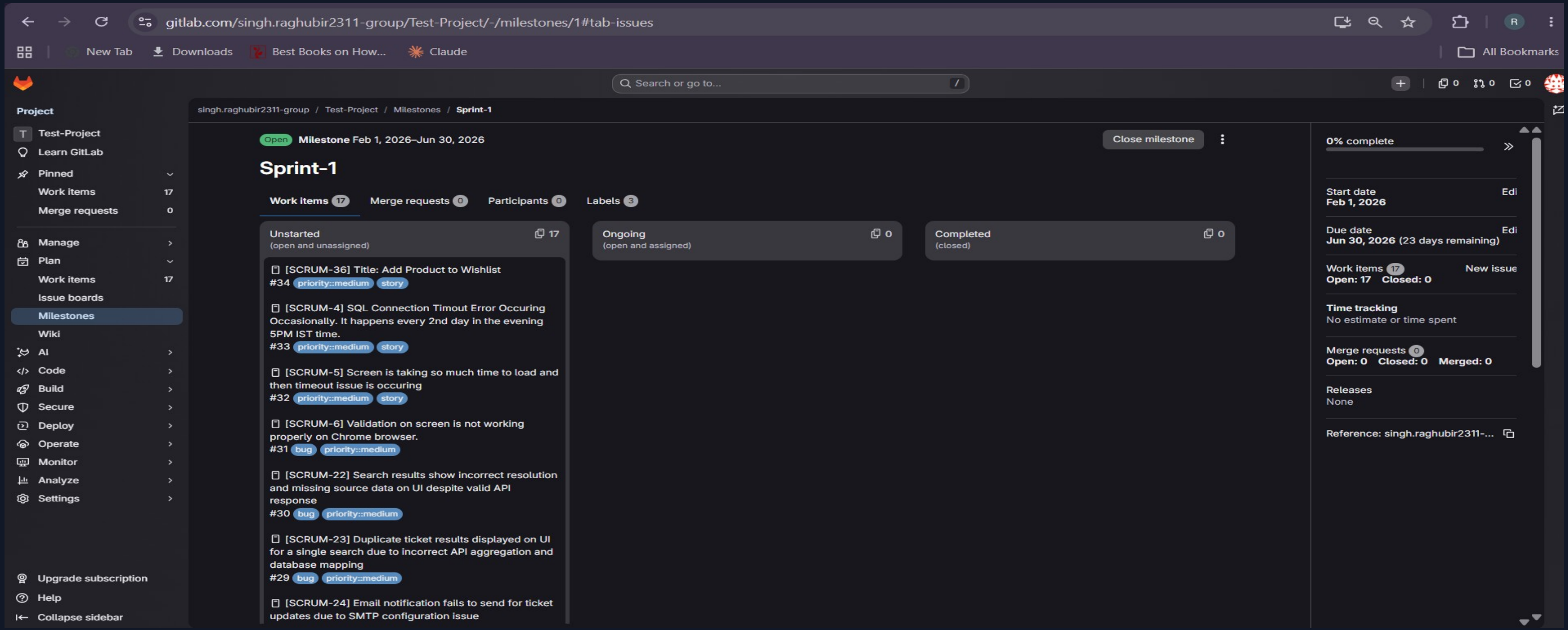


Result: Tickets Replicated to GitLab Sprint-1

17 Work Items

3 Labels

0 Failed



Authentication Setup

How the service authenticates with Jira and GitLab

🔑 Jira — API Token (Basic Auth)

01 Go to id.atlassian.com → Security → API tokens

02 Click Create API token — give it a name

03 Copy the token (shown only once)

04 Set Username + ApiToken in appsettings.json

05 Auth header: Basic base64(email:token)

🔒 GitLab — Personal Access Token

01 Go to gitlab.com/-/user_settings/personal_access_tokens

02 Click Add new token — name it JiraGitLabSync

03 Select scope: api (full read/write access)

04 Copy the generated token (starts with glpat-)

05 Set PrivateToken in appsettings.json

POC Complete

Jira → GitLab Sync Service

16+	100%	3-Level	0
Tickets Synced	Automated	Milestone Fallback	Manual Steps

Thank You • Questions Welcome